



ScienceDirect

Journal of Manufacturing Processes

Date: 30 May 2026

[Show more](#) ▼[Research article](#) [Full text access](#) [Get rights and content](#) ↗

Dynamics and stability of the keyhole in full-field power-modulated laser welding unveiled by in-situ X-ray imaging

[Chunkai Li](#) ^{a, b}  , [Zhongbo Zhao](#) ^a, [Yu Shi](#) ^{a, b}, [Yu Pan](#) ^a [Show more](#) ▼

View PDF

[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.jmapro.2026.03.027](#) ↗

Article

Highlights

- Revealed keyhole dynamics during full-field power-modulated galvanometer laser welding using in-situ microfocus real-time X-ray imaging.
- Key geometric features of the keyhole were extracted to quantitatively characterize its dynamic stability.
- Mechanisms of circular, infinity, and 8-shaped trajectories affecting keyhole stability and porosity suppression were elucidated.
- An energy density optimization for laser superposition regions improved keyhole stability and eliminated porosity.

Abstract

Full-field power-modulated laser welding is an emerging laser welding technology. It stabilizes the dynamic behavior of the keyhole by precisely regulating the spatiotemporal distribution of laser power. As a result, welding defects such as porosity and cracking are effectively suppressed. However, the large number of controllable process parameters, the difficulty in visualizing the keyhole, and the unclear mechanisms by which process parameters influence keyhole dynamics result in a lack of guidance in process optimization during practical welding applications. To address this issue, an in situ micro-focus X-ray real-time imaging system was used to directly monitor the dynamic behavior of the keyhole during laser welding of titanium alloy under different scanning trajectories and power modulation modes. Characteristic parameters, including depth variance, opening size, and depth-to-width ratio, were extracted. These parameters were combined with numerical simulation to analyze the effects of scanning trajectory and power modulation strategy on keyhole stability. The results indicate that, compared with conventional laser welding, the overall stability of the keyhole is significantly enhanced under circular, infinity, and 8-shaped scanning trajectory modes, accompanied by a pronounced reduction in weld porosity. However, a small number of pores are still observed in regions where the laser trajectories superposition. This phenomenon is attributed to a sudden increase in keyhole depth within the trajectory superposition regions, which leads to reduced stability and even keyhole collapse. By adopting a control strategy that lowers the energy density in the laser trajectory superposition regions, the dynamic stability of the keyhole can be further improved, which effectively suppresses welding porosity.

Keywords

Full-field power-modulated laser; Weld pool; Keyhole; In-situ X-ray imaging

[< Previous article in this issue](#)[Next article in this issue >](#)

Loading...

[View Abstract](#)

Recommended articles

Loading...

© 2026 The Society of Manufacturing Engineers. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

